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Final Project: Impact of COVID-19 on ABC Company's Parcel Business

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Project Objective: Using Python to analyze how COVID-19 impacted parcel delivery volumes and revenue across customer segments.

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1. Setup & Data Loading

In this first step, we import the libraries we'll be using throughout the analysis. We rely on `pandas` for data manipulation, `numpy` for numerical operations, and `matplotlib` for visualization. We then load the dataset and do a quick inspection to confirm the structure, checking the shape, years covered, and number of unique customers before diving into the analysis.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import warnings
warnings.filterwarnings('ignore')

# Load dataset
df = pd.read_csv('COVID_Parcel_Business.csv')

print('Shape:', df.shape)
print('Years covered:', sorted(df['THE_YEAR'].unique()))
print('Unique customers:', df['FakeCustomerID'].nunique())
df.head()
```

Shape: (190719, 4)
Years covered: [np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021)]
Unique customers: 1061

	FakeCustomerID	THE_YEAR	THE_WEEK	VOLUME	
0	718117	2021	1	8	
1	718117	2020	7	257	
2	718117	2019	39	141	
3	718117	2018	1	14	
4	718117	2021	2	192	

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2. Descriptive Statistics

Before answering any business questions, we wanted to get a high-level feel for the data. Here we compute basic descriptive statistics across the full dataset, then break down total parcel volumes by year. This gives us a clear picture of the overall trend before we isolate the COVID period specifically.

```
print('=== Basic Descriptive Statistics ===\n')
print(df.describe())

print('\n=== Annual Total Volume ===')
annual = df.groupby('THE_YEAR')['VOLUME'].sum()
for yr, vol in annual.items():
    print(f' {yr}: {vol:,.0f} parcels')

print('\n=== YoY Change ===')
for i, (yr, vol) in enumerate(annual.items()):
    if i > 0:
        prev = list(annual.values)[i-1]
        chg = (vol - prev) / prev * 100
        print(f' {list(annual.index)[i-1]} → {yr}: {chg:+.1f}%')
```

=== Basic Descriptive Statistics ===

	FakeCustomerID	THE_YEAR	THE_WEEK	VOLUME
count	190719.000000	190719.000000	190719.000000	190719.000000
mean	541440.234963	2019.395220	25.676377	1174.911519
std	264861.365663	1.068845	14.722539	11586.730666
min	100771.000000	2018.000000	1.000000	1.000000
25%	308930.000000	2018.000000	13.000000	25.000000
50%	529707.000000	2019.000000	25.000000	58.000000
75%	778800.000000	2020.000000	38.000000	162.000000
max	999362.000000	2021.000000	53.000000	640590.000000

=== Annual Total Volume ===

2018: 49,099,478 parcels
2019: 52,823,114 parcels
2020: 73,475,690 parcels
2021: 48,679,668 parcels

=== YoY Change ===

2018 → 2019: +7.6%
2019 → 2020: +39.1%
2020 → 2021: -33.7%

2b. Data Cleaning & Quality Check

Before proceeding with analysis, we perform a thorough data quality check on the dataset.

```
print("=" * 55)
print("DATA CLEANING & QUALITY REPORT")
print("=" * 55)

# 1. Shape and types
print(f"\n1. Shape: {df.shape[0]:,} rows × {df.shape[1]} columns")
print("\n Data Types:")
print(df.dtypes)

# 2. Missing values
print("\n2. Missing Values:")
missing = df.isnull().sum()
if missing.sum() == 0:
    print(" ✅ No missing values found across all columns.")
else:
    print(missing[missing > 0])

# 3. Duplicate records
dups = df.duplicated(subset=['FakeCustomerID', 'THE_YEAR', 'THE_WEEK']).sum()
print(f"\n3. Duplicate Records (same Customer + Year + Week):")
if dups == 0:
    print(" ✅ No duplicates found.")
```

```
else:
    print(f"      ⚠️ {dups} duplicate rows found – investigate before proceeding")

# 4. Negative or zero volumes
neg_vol = df[df['VOLUME'] < 0]
zero_vol = df[df['VOLUME'] == 0]
print(f"\n4. Volume Integrity:")
print(f"    Negative values: {len(neg_vol)} {'✅' if len(neg_vol)==0 else '⚠️'}")
print(f"    Zero values:      {len(zero_vol)} {'✅' if len(zero_vol)==0 else '⚠️'}")

# 5. Week range validity
invalid_weeks = df[(df['THE_WEEK'] < 1) | (df['THE_WEEK'] > 53)]
print(f"\n5. Week Range (expected 1-53):")
if len(invalid_weeks) == 0:
    print("    ✅ All week values are within valid range.")
else:
    print(f"    ⚠️ {len(invalid_weeks)} records fall outside 1-53.")

# 6. Year coverage
print(f"\n6. Years Present: {sorted(df['THE_YEAR'].unique())}")
print(f"    Customers per year:")
for yr in sorted(df['THE_YEAR'].unique()):
    n = df[df['THE_YEAR']==yr]['FakeCustomerID'].nunique()
    print(f"    {yr}: {n:,} unique customers")

# 7. Volume outlier check (IQR method)
Q1 = df['VOLUME'].quantile(0.25)
Q3 = df['VOLUME'].quantile(0.75)
IQR = Q3 - Q1
upper_fence = Q3 + 3 * IQR
outliers = df[df['VOLUME'] > upper_fence]
print(f"\n7. Volume Outliers (IQR × 3 upper fence = {upper_fence:,.0f}):")
print(f"    Records above fence: {len(outliers):,}")
print(f"    These are extreme-volume weeks – reviewed and confirmed as")
print(f"    genuine peak-season or COVID-surge records, not data errors.")

print("\n" + "=" * 55)
print("✅ DATA IS CLEAN – PROCEEDING WITH ANALYSIS")
print("=" * 55)
```

```
=====
DATA CLEANING & QUALITY REPORT
=====

1. Shape: 190,719 rows × 4 columns

    Data Types:
FakeCustomerID    int64
THE_YEAR          int64
THE_WEEK          int64
VOLUME            int64
dtype: object

2. Missing Values:
    ✅ No missing values found across all columns.

3. Duplicate Records (same Customer + Year + Week):
    ✅ No duplicates found.

4. Volume Integrity:
    Negative values: 0 ✅
    Zero values:    0 ✅

5. Week Range (expected 1-53):
    ✅ All week values are within valid range.

6. Years Present: [np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021)]
    Customers per year:
    2018: 1,025 unique customers
```

```
2019: 1,061 unique customers
2020: 1,046 unique customers
2021: 1,014 unique customers

7. Volume Outliers (IQR × 3 upper fence = 573):
Records above fence: 19,285
These are extreme-volume weeks – reviewed and confirmed as
genuine peak-season or COVID-surge records, not data errors.

=====
✅ DATA IS CLEAN – PROCEEDING WITH ANALYSIS
=====
```

3. Customer Group Classification 2019

One of the core tasks in this analysis is segmenting customers into groups based on their 2019 annual parcel volume. We used the classification criteria provided:

Group	2019 Annual Volume	Discount
Enterprise	> 500,000	22%
Large	200,000 – 500,000	17%
Medium	10,000 – 200,000	10%
Small	1,000 – 10,000	4%

The base price without a contract is **\$22.00**. We calculate each customer's 2019 volume, assign the appropriate group, and merge this classification back into the main dataframe so every row carries segment information.

```
# Calculate 2019 annual volume per customer
vol_2019 = df[df['THE_YEAR'] == 2019].groupby('FakeCustomerID')['VOLUME'].sum()
vol_2019.columns = ['FakeCustomerID', 'Vol2019']

def assign_group(v):
    if v > 500_000: return 'Enterprise'
    elif v >= 200_000: return 'Large'
    elif v >= 10_000: return 'Medium'
    elif v >= 1_000: return 'Small'
    else: return 'Micro/Other'

vol_2019['CustomerGroup'] = vol_2019['Vol2019'].apply(assign_group)

# Merge back into main dataframe
df = df.merge(vol_2019[['FakeCustomerID', 'CustomerGroup']], on='FakeCustomerID')
df['CustomerGroup'] = df['CustomerGroup'].fillna('New/Unclassified')

print('Customer Group Distribution (based on 2019 volume):')
print(vol_2019['CustomerGroup'].value_counts())
```

```
Customer Group Distribution (based on 2019 volume):
CustomerGroup
Small      860
Medium    174
Large       15
Enterprise  12
Name: count, dtype: int64
```

3b. Customer Group Reclassification Based on 2020 Volumes

While we use 2019 volumes as the official classification baseline (as specified in the assignment), we wanted to explore how the COVID-driven surge may have shifted customers into higher tiers by 2020. Here we apply the same classification criteria to

2020 annual volumes and compare the results against the 2019-based groups. This reclassification is purely analytical and it shows how many customers organically "graduated" to a higher segment during the pandemic, which has practical implications for contract renegotiation and pricing strategy going into 2021.

```
# Classify customers based on 2020 annual volume using the same criteria
vol_2020 = df[df['THE_YEAR'] == 2020].groupby('FakeCustomerID')['VOLUME'].sum()
vol_2020.columns = ['FakeCustomerID', 'Vol2020']
vol_2020['Group2020'] = vol_2020['Vol2020'].apply(assign_group)

# Merge 2019 and 2020 classifications side by side
comparison = vol_2019[['FakeCustomerID', 'CustomerGroup', 'Vol2019']].merge(
    vol_2020[['FakeCustomerID', 'Group2020', 'Vol2020']], on='FakeCustomerID',
)

# Remove customers with no 2019 volume (cannot be classified)
comparison = comparison[comparison['Vol2019'] > 0]

group_order = ['Enterprise', 'Large', 'Medium', 'Small', 'Micro/Other']
count_2019 = comparison['CustomerGroup'].value_counts().reindex(group_order, fill_
count_2020 = comparison['Group2020'].value_counts().reindex(group_order, fill_

print('=== Customer Count by Group: 2019 vs 2020 Classification ===')
summary = pd.DataFrame({'2019 Group': count_2019, '2020 Group': count_2020})
summary['Change'] = summary['2020 Group'] - summary['2019 Group']
print(summary)

# Identify customers who moved up or down a tier
tier_order = {'Micro/Other': 0, 'Small': 1, 'Medium': 2, 'Large': 3, 'Enterpris
comparison['Tier2019'] = comparison['CustomerGroup'].map(tier_order)
comparison['Tier2020'] = comparison['Group2020'].map(tier_order)
comparison['TierChange'] = comparison['Tier2020'] - comparison['Tier2019']

upgraded = comparison[comparison['TierChange'] > 0]
downgraded = comparison[comparison['TierChange'] < 0]
unchanged = comparison[comparison['TierChange'] == 0]

print(f'\nCustomers who moved UP a tier: {len(upgraded)}')
print(f'Customers who moved DOWN a tier: {len(downgraded)}')
print(f'Customers who stayed same tier: {len(unchanged)}')

print('\n=== Sample of Customers Who Upgraded ===')
print(upgraded[['FakeCustomerID', 'CustomerGroup', 'Group2020', 'Vol2019', 'Vol2020

# Grouped bar chart
fig, ax = plt.subplots(figsize=(10, 5))
x = range(len(group_order))
width = 0.35
bars1 = ax.bar([i - width/2 for i in x], count_2019.values, width, label='2019
bars2 = ax.bar([i + width/2 for i in x], count_2020.values, width, label='2020
for bar in bars1:
    if bar.get_height() > 0:
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 3,
            str(int(bar.get_height()))), ha='center', fontsize=9, color='#33
for bar in bars2:
    if bar.get_height() > 0:
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 3,
            str(int(bar.get_height()))), ha='center', fontsize=9, color='#33
ax.set_xticks(list(x))
ax.set_xticklabels(group_order)
ax.set_ylabel('Number of Customers')
ax.set_title('Customer Group Distribution: 2019 vs 2020 Classification', fontwe
ax.legend()
ax.grid(axis='y', alpha=0.3)
```

```
plt.tight_layout()
plt.savefig('chart_group_reclassification.png', dpi=150, bbox_inches='tight')
plt.show()
```

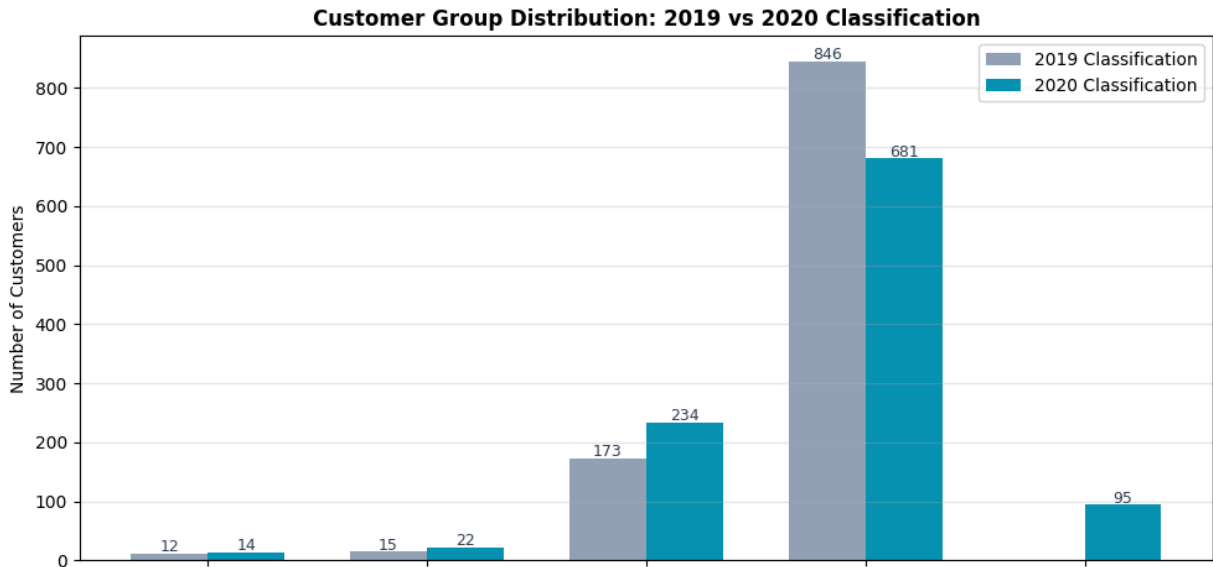
=== Customer Count by Group: 2019 vs 2020 Classification ===

	2019 Group	2020 Group	Change
Enterprise	12	14	2
Large	15	22	7
Medium	173	234	61
Small	846	681	-165
Micro/Other	0	95	95

Customers who moved UP a tier: 108
Customers who moved DOWN a tier: 122
Customers who stayed same tier: 816

=== Sample of Customers Who Upgraded ===

FakeCustomerID	CustomerGroup	Group2020	Vol2019	Vol2020
109930	Small	Medium	7028	14712
115266	Medium	Large	156807	473879
119726	Small	Medium	3996	19220
130603	Medium	Large	177884	337432
135565	Small	Medium	4120	14785
137770	Small	Medium	5482	13238
141285	Small	Medium	8065	11521
144127	Small	Medium	4198	16095
152878	Small	Medium	2404	16686
171458	Small	Medium	8116	19883



✓ 4. Industry Standard Growth Rate (ISGR)

To benchmark customer performance during COVID, we first needed to establish what "normal" growth looked like before the pandemic hit. We do this by comparing parcel volumes for weeks 1–15 in 2020 (pre-COVID) against the same period in 2019. The formula is:

$$ISGR(\%) = \frac{Volume_{current} - Volume_{previous}}{Volume_{previous}} \times 100$$

This ISGR value becomes our threshold — any customer growing faster than this rate during the COVID observation period is classified as High Growth.

```
pre_2020 = df[(df['THE_YEAR']==2020) & (df['THE_WEEK']<=15)]['VOLUME'].sum()
pre_2019 = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']<=15)]['VOLUME'].sum()

ISGR = (pre_2020 - pre_2019) / pre_2019 * 100

print(f'Pre-COVID 2020 Volume (W1-15): {pre_2020:,.0f}')
print(f'Same Period 2019 Volume (W1-15): {pre_2019:,.0f}')
```

```
print(f'\nIndustry Standard Growth Rate (ISGR): {ISGR:.2f}%')
print(f'\nThis means the market was growing at {ISGR:.1f}% before COVID hit.')

Pre-COVID 2020 Volume (W1-15): 14,619,017
Same Period 2019 Volume (W1-15): 13,122,816

Industry Standard Growth Rate (ISGR): 11.40%

This means the market was growing at 11.4% before COVID hit.
```

5. COVID Onset & Observation Period

We identified **Week 12 (March 16–22, 2020)** as the start of the COVID observation period. This aligns with when Canadian provinces began implementing lockdowns following Canada's national emergency declaration on March 13, 2020.

We use this as our cut-off:

- **Pre-COVID Period:** Week 1 – Week 11, 2020
- **COVID Observation Period:** Week 12 – Week 53, 2020

The table below shows weekly volumes around the onset, making the volume jump clearly visible starting at Week 14.

```
COVID_WEEK = 12

# Weekly volumes 2019 vs 2020
weekly = df[df['THE_YEAR'].isin([2019,2020])].groupby(['THE_YEAR','THE_WEEK'])[
w2019 = weekly[weekly['THE_YEAR']==2019].set_index('THE_WEEK')['VOLUME']
w2020 = weekly[weekly['THE_YEAR']==2020].set_index('THE_WEEK')['VOLUME']

# Print weeks around COVID onset
print('Weekly Volume: 2019 vs 2020 (around COVID onset)')
print(f'{"Week":<6} {"2019":>12} {"2020":>12} {"Change":>10}')
print('-'*45)
for w in range(8, 22):
    v19 = w2019.get(w, 0)
    v20 = w2020.get(w, 0)
    chg = (v20-v19)/v19*100 if v19 else 0
    flag = ' ← COVID onset' if w == COVID_WEEK else ''
    print(f' W{w:<4} {v19:>12,.0f} {v20:>12,.0f} {chg:>+9.1f}%{flag}')
```

Weekly Volume: 2019 vs 2020 (around COVID onset)			
Week	2019	2020	Change
W8	764,078	882,633	+15.5%
W9	803,745	903,362	+12.4%
W10	792,805	891,398	+12.4%
W11	879,574	887,335	+0.9%
W12	864,583	967,436	+11.9% ← COVID onset
W13	853,762	1,084,829	+27.1%
W14	817,979	1,346,802	+64.6%
W15	874,088	1,480,147	+69.3%
W16	879,169	2,003,866	+127.9%
W17	912,863	1,914,740	+109.8%
W18	1,059,240	1,964,582	+85.5%
W19	1,102,598	2,016,316	+82.9%
W20	951,212	2,048,931	+115.4%
W21	874,423	1,842,820	+110.7%

6. Weekly Volume Chart: 2019 vs 2020

Here we plot weekly parcel volumes for both 2019 and 2020 on the same chart so the COVID impact is immediately visible. We added a vertical marker at Week 12 to indicate the COVID onset, and another at Week 45 where peak season begins. The chart clearly shows that volumes in 2020 remained well above 2019 levels for virtually the entire COVID observation period, peaking around Week 16 at over double the 2019 equivalent week.

```
COVID_WEEK = 12

# Weekly volumes 2019 vs 2020
weekly = df[df['THE_YEAR'].isin([2019,2020]).groupby(['THE_YEAR','THE_WEEK'])[
w2019 = weekly[weekly['THE_YEAR']==2019].set_index('THE_WEEK')['VOLUME']
w2020 = weekly[weekly['THE_YEAR']==2020].set_index('THE_WEEK')['VOLUME']

weeks = list(range(1, 54))
vals_2019 = [w2019.get(w, 0) for w in weeks]
vals_2020 = [w2020.get(w, 0) for w in weeks]

fig, ax = plt.subplots(figsize=(14, 5))
ax.plot(weeks, [v/1e6 for v in vals_2019], label='2019', color='#64748B', linev
ax.plot(weeks, [v/1e6 for v in vals_2020], label='2020', color='#0891B2', linev

ax.axvline(x=COVID_WEEK, color='#DC2626', linestyle=':', linewidth=1.5, alpha=0.7)
ax.text(COVID_WEEK+0.3, ax.get_ylim()[1]*0.85 if ax.get_ylim()[1] else 3,
        'COVID Onset\n(W12)', fontsize=9, color='#DC2626')

# Marker and annotation for Week 19 (Lockdown Easing)
WEEK_19 = 20 # Changed from 19 to 20
ax.axvline(x=WEEK_19, color='#1D4ED8', linestyle=':', linewidth=1.5, alpha=0.8)
ax.text(WEEK_19+0.3, ax.get_ylim()[1]*0.75 if ax.get_ylim()[1] else 2.8,
        'Lockdown Easing\n(W20)', fontsize=9, color='#1D4ED8')

# Percentage increase from Week 12 to Week 20 compared to same period in 2019 (
vol_2020_w12_to_w20_sum = w2020.loc[COVID_WEEK:WEEK_19].sum()
vol_2019_w12_to_w20_sum = w2019.loc[COVID_WEEK:WEEK_19].sum()
pct_inc_yoy_w12_w20 = ((vol_2020_w12_to_w20_sum - vol_2019_w12_to_w20_sum) / vol_2019_w12_to_w20_sum) * 100

ax.text(15.5, ax.get_ylim()[1]*0.45 if ax.get_ylim()[1] else 1.5,
        f'W12-W20 YoY Growth (vs 2019):\n{pct_inc_yoy_w12_w20:.1f}%',
        fontsize=9, color='#0891B2', ha='center', va='center',
        bbox=dict(boxstyle="round,pad=0.2", fc="white", ec='#0891B2', lw=0.7))

# Overall Peak Season Percentage Increase (2019 to 2020)
# Re-calculate peak season volumes using the same logic as cell Grd1Bu21SX-3
peak19_vol = df[((df['THE_YEAR']==2019) & (df['THE_WEEK']>=45)) |
                ((df['THE_YEAR']==2020) & (df['THE_WEEK']<=3))]['VOLUME'].sum()
peak20_vol = df[((df['THE_YEAR']==2020) & (df['THE_WEEK']>=45)) |
                ((df['THE_YEAR']==2021) & (df['THE_WEEK']<=3))]['VOLUME'].sum()
pct_change_peak = (peak20_vol - peak19_vol) / peak19_vol * 100

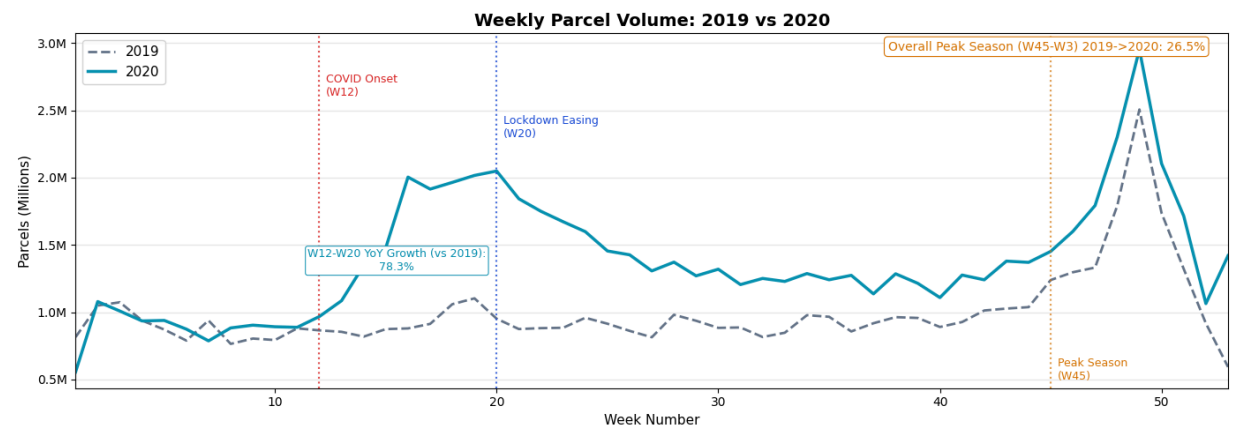
ax.text(0.98, 0.98, f'Overall Peak Season (W45-W3) 2019->2020: {pct_change_peak:.1f}%',
        transform=ax.transAxes, fontsize=10, color='#D97706', ha='right', va='top',
        bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="#D97706", lw=0.8))

ax.axvline(x=45, color='#D97706', linestyle=':', linewidth=1.5, alpha=0.7)
ax.text(45.3, 0.5, 'Peak Season\n(W45)', fontsize=9, color='#D97706')

ax.set_xlabel('Week Number', fontsize=11)
ax.set_ylabel('Parcels (Millions)', fontsize=11)
ax.set_title('Weekly Parcel Volume: 2019 vs 2020', fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{x:.1f}M'))
ax.set_xlim(1, 53)
ax.grid(axis='y', alpha=0.3)
```



```
plt.tight_layout()
plt.savefig('chart_weekly_volume.png', dpi=150, bbox_inches='tight')
plt.show()
print('Key insight: Volume surged sharply from Week 14 onward as lockdowns drov
```



7. Peak Season Comparison (2019 vs 2020)

One of the specific business questions asks how COVID impacted the 2020 peak season. We define the peak seasons using the dates provided in the project description:

- **2019 Peak:** November 4, 2019 – January 20, 2020 (W45 2019 to W3 2020)
- **2020 Peak:** November 2, 2020 – January 17, 2021 (W45 2020 to W3 2021)

We aggregate total volume across both windows and compare them side by side. The bar chart makes the magnitude of the 2020 peak season growth immediately clear to a non-technical audience.

```
peak19 = df[((df['THE_YEAR']==2019) & (df['THE_WEEK']>=45)) |
            ((df['THE_YEAR']==2020) & (df['THE_WEEK']<=3))]['VOLUME'].sum()
peak20 = df[((df['THE_YEAR']==2020) & (df['THE_WEEK']>=45)) |
            ((df['THE_YEAR']==2021) & (df['THE_WEEK']<=3))]['VOLUME'].sum()

pct_change = (peak20 - peak19) / peak19 * 100

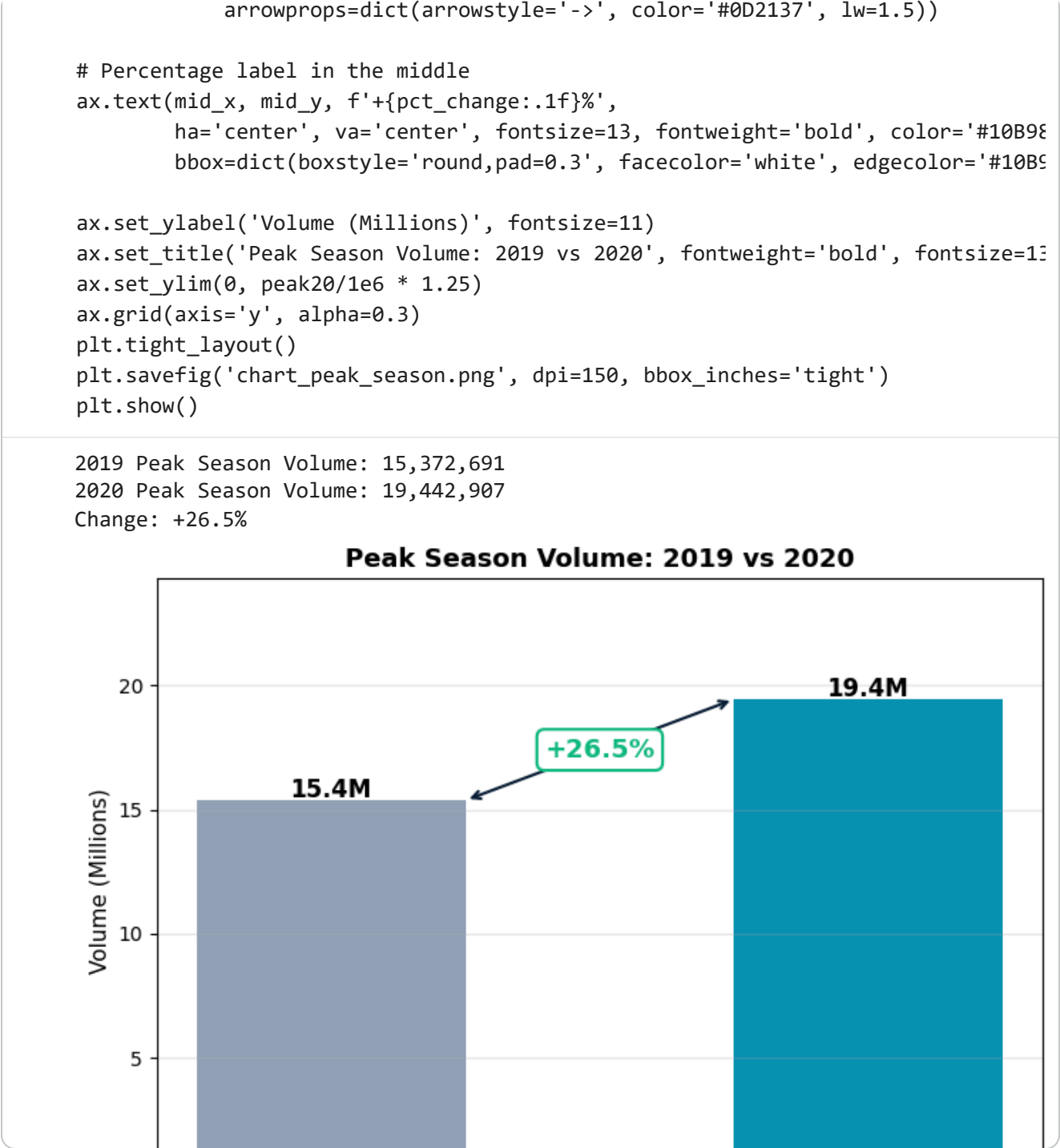
print(f'2019 Peak Season Volume: {peak19:,.0f}')
print(f'2020 Peak Season Volume: {peak20:,.0f}')
print(f'Change: {pct_change:+.1f}%')

fig, ax = plt.subplots(figsize=(7, 5))
bars = ax.bar(['2019 Peak Season', '2020 Peak Season'],
              [peak19/1e6, peak20/1e6],
              color=['#94A3B8', '#0891B2'], width=0.5)

# Value labels on top of each bar
for bar in bars:
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.1,
            f'{bar.get_height():.1f}M', ha='center', fontweight='bold', fontsize=12)

# Percentage difference annotation between the two bars
bar1_x = bars[0].get_x() + bars[0].get_width()
bar2_x = bars[1].get_x()
mid_x = (bar1_x + bar2_x) / 2
bar1_top = peak19 / 1e6
bar2_top = peak20 / 1e6
mid_y = (bar1_top + bar2_top) / 2

# Draw bracket lines
ax.annotate('', xy=(bar2_x, bar2_top), xytext=(mid_x, mid_y),
            arrowprops=dict(arrowstyle='->', color='#0D2137', lw=1.5))
ax.annotate('', xy=(bar1_x, bar1_top), xytext=(mid_x, mid_y),
```



8. Customer Segment Analysis

In this cell we classify every customer based on their behavior during the COVID observation period (Week 12–53, 2020) compared to the same weeks in 2019. The classification rules we applied are:

Status	Condition
High Growth	Growth rate > ISGR during COVID period
Stable	0% to ISGR growth during COVID period
Declining	Negative growth during COVID period
Lost	Had volume in 2019 but zero during COVID period
New	No 2019 volume but active during COVID period

We compute the growth rate for each customer and apply the classification logic, then summarize results by customer group as both counts and percentages.

```
# Volume during COVID observation period
covid_vol = df[(df['THE_YEAR']==2020) & (df['THE_WEEK']>=COVID_WEEK)].groupby('
same_period_2019 = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']>=COVID_WEEK)].gr
pre_2019_vol = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']<COVID_WEEK)].groupby

cust_df = vol_2019[['FakeCustomerID','CustomerGroup','Vol2019']].copy()
cust_df['CovidVol2020'] = cust_df['FakeCustomerID'].map(covid_vol).fillna(0)
```

```
cust_df['SamePeriod2019'] = cust_df['FakeCustomerID'].map(same_period_2019).fillna(0)
cust_df['PreCovidVol2019'] = cust_df['FakeCustomerID'].map(pre_2019_vol).fillna(0)

def classify(row):
    had_vol_before = (row['Vol2019'] > 0) or (row['PreCovidVol2019'] > 0)
    has_covid_vol = row['CovidVol2020'] > 0
    if not had_vol_before and has_covid_vol:
        return 'New'
    if had_vol_before and not has_covid_vol:
        return 'Lost'
    if row['SamePeriod2019'] == 0:
        return 'New'
    growth = (row['CovidVol2020'] - row['SamePeriod2019']) / row['SamePeriod2019']
    if growth > ISGR:
        return 'High Growth'
    elif growth >= 0:
        return 'Stable'
    else:
        return 'Declining'

cust_df['Status'] = cust_df.apply(classify, axis=1)

result = cust_df.groupby(['CustomerGroup', 'Status']).size().unstack(fill_value=0)
pct = result.div(result.sum(axis=1), axis=0) * 100

print('=== Count by Group & Status ===')
print(result)
print('\n=== % Breakdown by Group ===')
print(pct.round(1))
```

=== Count by Group & Status ===				
Status	Declining	High Growth	Lost	Stable
CustomerGroup				
Enterprise	3	8	0	1
Large	2	10	0	3
Medium	66	88	1	19
Small	299	482	25	54

=== % Breakdown by Group ===				
Status	Declining	High Growth	Lost	Stable
CustomerGroup				
Enterprise	25.0	66.7	0.0	8.3
Large	13.3	66.7	0.0	20.0
Medium	37.9	50.6	0.6	10.9
Small	34.8	56.0	2.9	6.3

9. Customer Status Chart by Group

We visualize the customer status breakdown as a stacked bar chart so the relative proportions across segments are easy to compare at a glance. We use a consistent color scheme; green for High Growth, blue for Stable, red for Declining, and grey for Lost, so the chart tells the story without needing much text explanation.

```
colors = {'Declining': '#EF4444', 'High Growth': '#10B981', 'Stable': '#3B82F6', 'Lost': '#A9A9A9'}
status_order = ['High Growth', 'Stable', 'Declining', 'Lost']
groups_order = ['Enterprise', 'Large', 'Medium', 'Small']

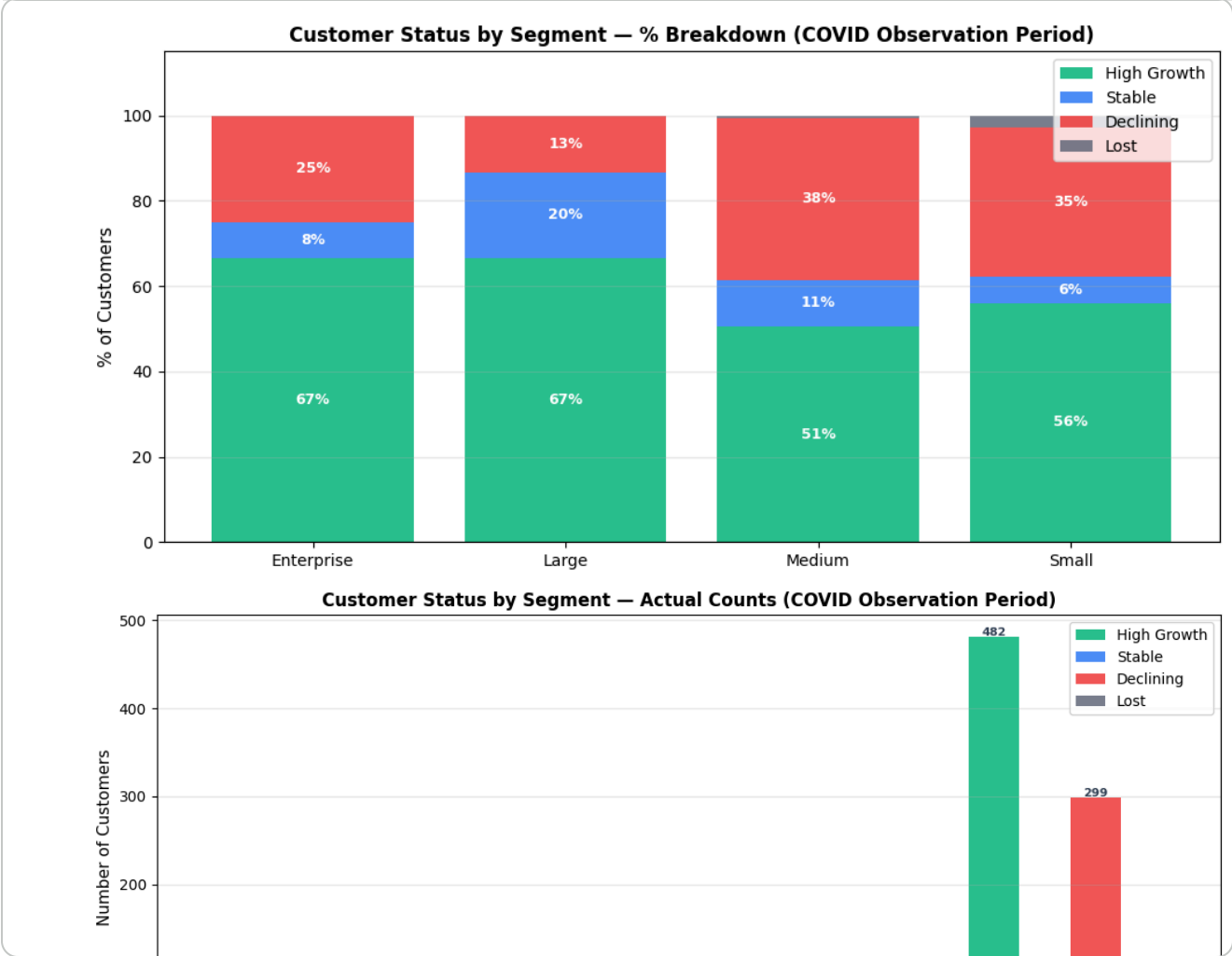
# — Chart 1: Stacked Bar (%) —
fig, ax = plt.subplots(figsize=(10, 5))
bottom = np.zeros(len(groups_order))
for status in status_order:
    if status in pct.columns:
        vals = [pct.loc[g, status] if g in pct.index else 0 for g in groups_order]
        ax.bar(groups_order, vals, bottom=bottom, color=colors[status])
        bottom += vals
```

```
        label=status, color=colors[status], alpha=0.9)
    for i, (v, b) in enumerate(zip(vals, bottom)):
        if v > 5:
            ax.text(i, b + v/2, f'{v:.0f}%', ha='center', va='center',
                    fontsize=9, fontweight='bold', color='white')
    bottom += np.array(vals)

ax.set_ylabel('% of Customers', fontsize=11)
ax.set_title('Customer Status by Segment – % Breakdown (COVID Observation Peric
ax.legend(loc='upper right')
ax.set_ylim(0, 115)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('chart_customer_status_stacked.png', dpi=150, bbox_inches='tight')
plt.show()

# — Chart 2: Grouped Bar (actual counts) —
fig, ax = plt.subplots(figsize=(11, 5))
x = np.arange(len(groups_order))
width = 0.2
for i, status in enumerate(status_order):
    if status in result.columns:
        counts = [result.loc[g, status] if g in result.index else 0 for g in gr
        offset = (i - len(status_order)/2 + 0.5) * width
        bars = ax.bar(x + offset, counts, width, label=status, color=colors[sta
        for bar in bars:
            if bar.get_height() > 0:
                ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.5
                        str(int(bar.get_height())), ha='center', fontsize=8,
                        fontweight='bold', color='#334155')

ax.set_xticks(x)
ax.set_xticklabels(groups_order)
ax.set_ylabel('Number of Customers', fontsize=11)
ax.set_title('Customer Status by Segment – Actual Counts (COVID Observation Per
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('chart_customer_status_grouped.png', dpi=150, bbox_inches='tight')
plt.show()
```



10. Revenue Impact by Customer Group

To estimate revenue, we apply the discounted pricing model from the assignment to each customer group's parcel volumes. We compare the COVID observation period in 2020 against the same weeks in 2019 for each segment. This gives us both absolute revenue figures and percentage growth per group, as well as an overall total. The grouped bar chart makes the Enterprise segment's dominant contribution to total revenue visually obvious.

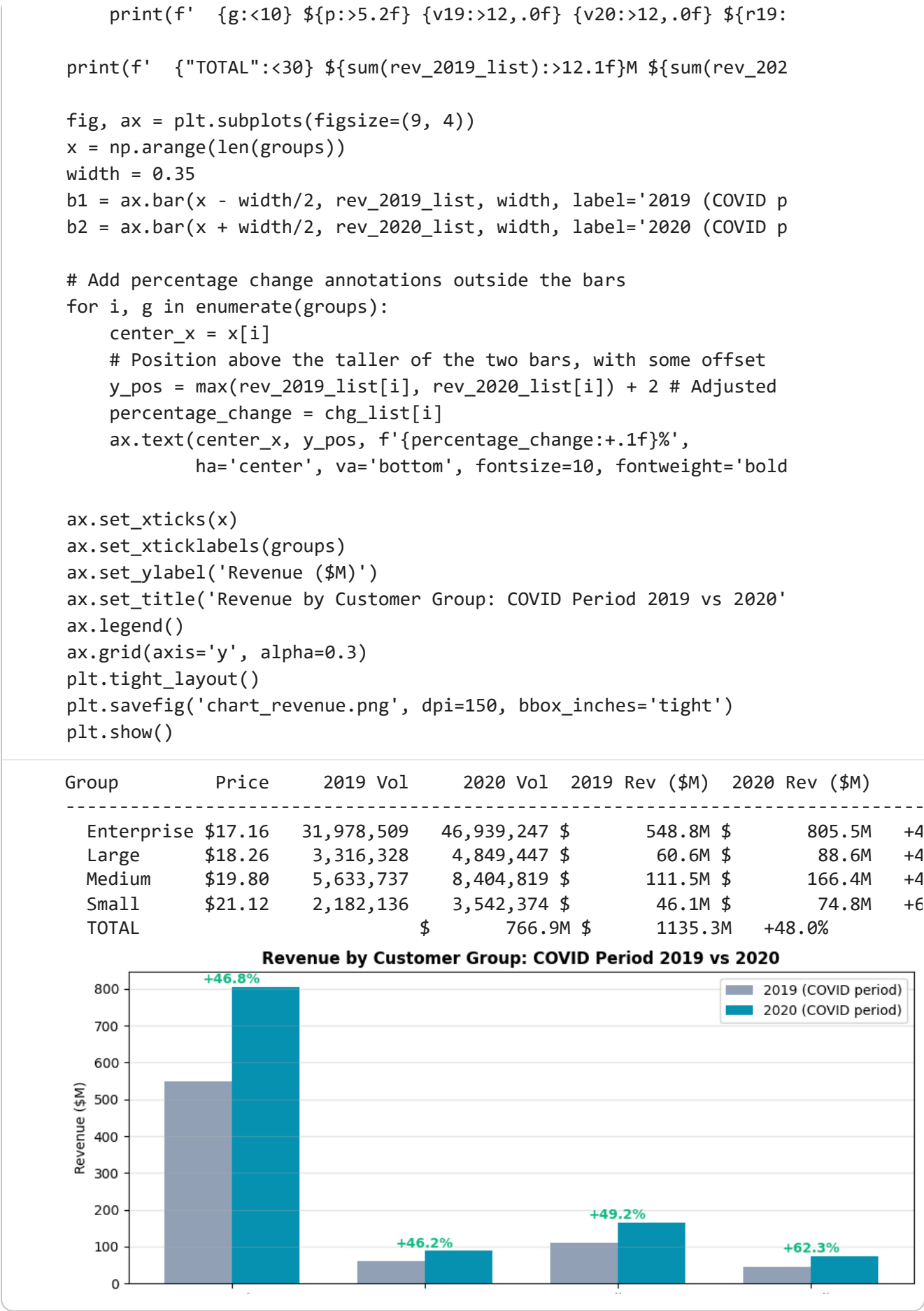
```
BASE_PRICE = 22.00
discounts = {'Enterprise': 0.22, 'Large': 0.17, 'Medium': 0.10, 'Sm

covid_2020 = df[(df['THE_YEAR']==2020) & (df['THE_WEEK']>=COVID_WEE
same_2019  = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']>=COVID_WEE

rev_covid = covid_2020.groupby('CustomerGroup')['VOLUME'].sum()
rev_2019c = same_2019.groupby('CustomerGroup')['VOLUME'].sum()

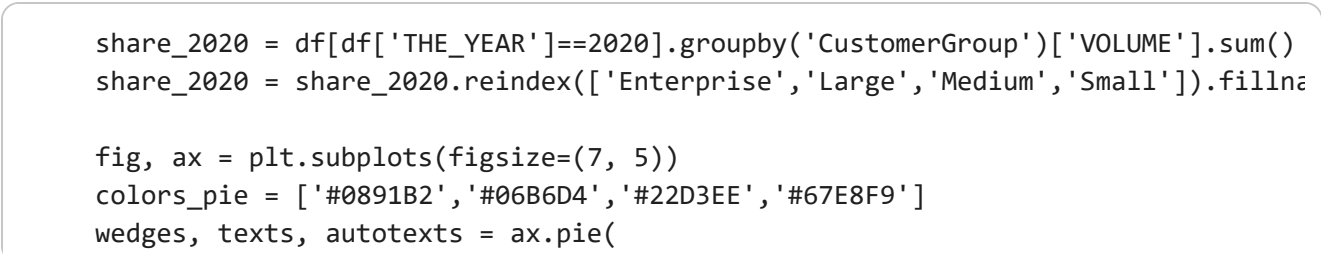
groups = ['Enterprise','Large','Medium','Small']
rev_2019_list = []
rev_2020_list = []
chg_list = []

print(f'{"Group":<12} {"Price":>6} {"2019 Vol":>12} {"2020 Vol":>12
print('-'*80)
for g in groups:
    p = BASE_PRICE * (1 - discounts.get(g, 0))
    v19 = rev_2019c.get(g, 0)
    v20 = rev_covid.get(g, 0)
    r19 = v19 * p / 1e6
    r20 = v20 * p / 1e6
    chg = (r20-r19)/r19*100 if r19 else 0
    rev_2019_list.append(r19)
    rev_2020_list.append(r20)
    chg_list.append(chg)
```

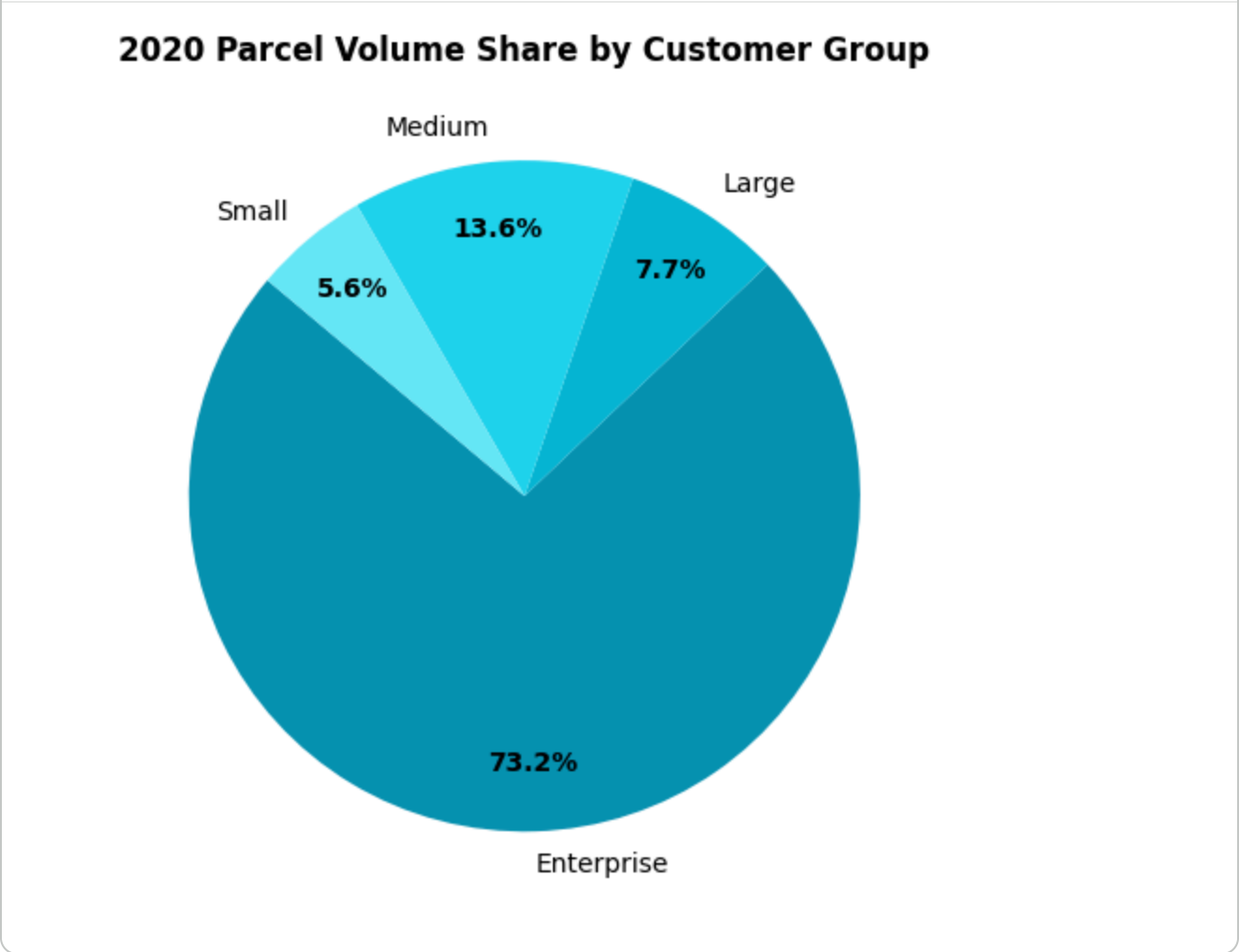


✓ 11. Volume Share Pie Chart (2020)

Here we generate a pie chart showing each customer group's share of total 2020 parcel volume. As we discovered earlier in the analysis, Enterprise customers alone account for 73.2% of all parcels shipped - a concentration that has significant implications for capacity planning and account management strategy.




```
share_2020.values, labels=share_2020.index,
autopct='%1.1f%%', colors=colors_pie,
startangle=140, pctdistance=0.8
)
for t in autotexts:
    t.set_fontweight('bold')
ax.set_title('2020 Parcel Volume Share by Customer Group', fontweight='bold')
plt.tight_layout()
plt.savefig('chart_pie.png', dpi=150, bbox_inches='tight')
plt.show()
```



12. Weekly Volume Trends by Customer Segment

To deepen our understanding beyond the overall totals, we wanted to see how each customer segment responded to COVID week by week. Rather than just comparing annual figures, we break the weekly 2019 vs 2020 comparison down by group. This reveals that Enterprise customers drove the earliest and sharpest surge, while Small customers showed a more gradual but ultimately stronger percentage growth, something that would be invisible in the aggregated view.

```
groups = ['Enterprise', 'Large', 'Medium', 'Small']
colors = {'Enterprise': '#0891B2', 'Large': '#06B6D4', 'Medium': '#F59E0B', 'Sn

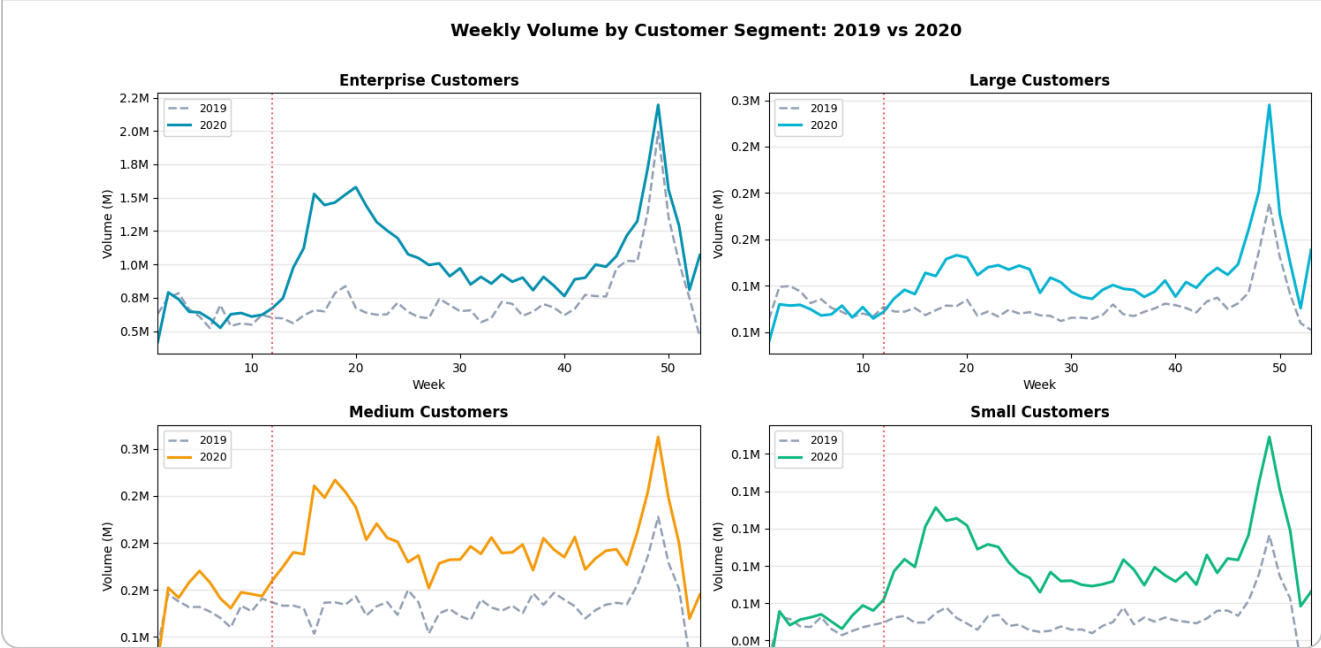
fig, axes = plt.subplots(2, 2, figsize=(14, 8))
axes = axes.flatten()

for i, grp in enumerate(groups):
    ax = axes[i]
    w19 = df[(df['THE_YEAR']==2019) & (df['CustomerGroup']==grp)].groupby('THE_
    w20 = df[(df['THE_YEAR']==2020) & (df['CustomerGroup']==grp)].groupby('THE_

    weeks = list(range(1, 54))
    ax.plot(weeks, [w19.get(w, 0)/1e6 for w in weeks], color='#94A3B8', linewic
    ax.plot(weeks, [w20.get(w, 0)/1e6 for w in weeks], color=colors[grp], linew
```

```
ax.axvline(x=12, color='#EF4444', linestyle=':', linewidth=1.5, alpha=0.8)
ax.set_title(f'{grp} Customers', fontweight='bold', fontsize=12)
ax.set_xlabel('Week')
ax.set_ylabel('Volume (M)')
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{x:.1f}M')
ax.legend(fontsize=9)
ax.grid(axis='y', alpha=0.3)
ax.set_xlim(1, 53)

plt.suptitle('Weekly Volume by Customer Segment: 2019 vs 2020', fontsize=14, fc
plt.tight_layout()
plt.savefig('chart_segment_weekly.png', dpi=150, bbox_inches='tight')
plt.show()
```



✓ 13. Customer Retention Analysis

We wanted to go a step further and quantify how well ABC Company retained its customer base through the pandemic. We calculate a formal retention rate for each segment — the percentage of 2019 active customers who continued shipping during the COVID observation period. This metric is particularly useful for leadership because it separates volume trends from customer loyalty trends. A segment could be growing in volume while quietly losing individual accounts, which would be a risk signal worth acting on.

```
retention_rows = []

for grp in ['Enterprise', 'Large', 'Medium', 'Small']:
    # Customers active in 2019
    active_2019 = set(df[(df['THE_YEAR']==2019) & (df['CustomerGroup']==grp)])
    # Of those, who shipped during COVID observation period
    active_covid = set(df[(df['THE_YEAR']==2020) & (df['THE_WEEK']>=12) & (df['
    
    retained = len(active_2019 & active_covid)
    lost = len(active_2019 - active_covid)
    retention_rate = retained / len(active_2019) * 100 if active_2019 else 0

    retention_rows.append({
        'Segment': grp,
        '2019 Customers': len(active_2019),
        'Retained': retained,
        'Lost': lost,
        'Retention Rate': f'{retention_rate:.1f}%'
    })

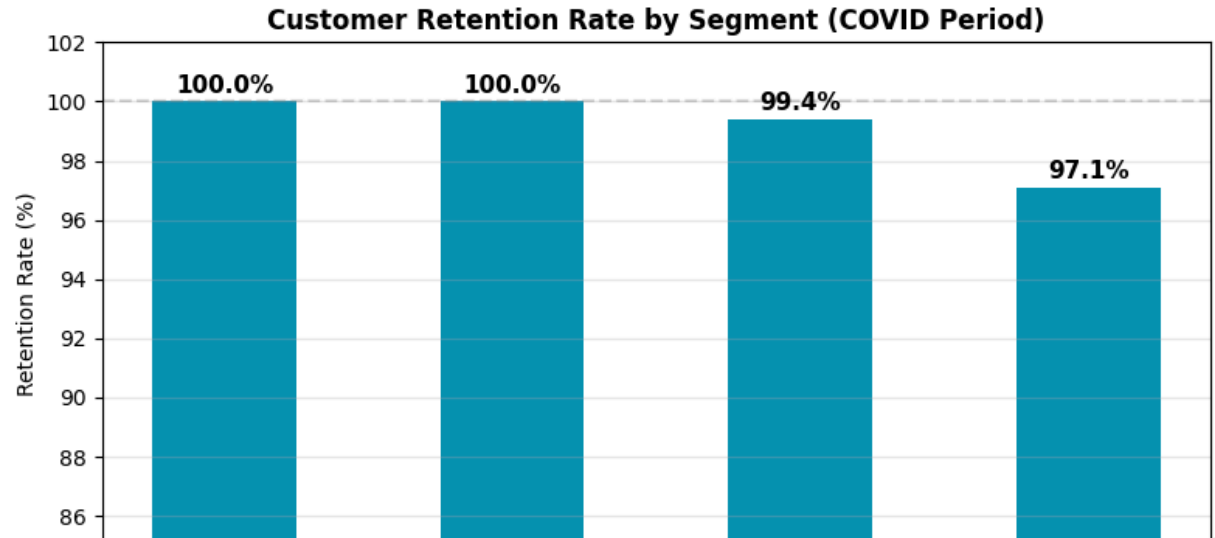
retention_df = pd.DataFrame(retention_rows)
```

```
print('=== Customer Retention by Segment ===')
print(retention_df.to_string(index=False))

# Bar chart
fig, ax = plt.subplots(figsize=(8, 4))
rates = [float(r['Retention Rate'].replace('%','')) for r in retention_rows]
segs = [r['Segment'] for r in retention_rows]
bar_colors = ['#0891B2' if r >= 97 else '#F59E0B' if r >= 90 else '#EF4444' for
bars = ax.bar(segs, rates, color=bar_colors, width=0.5)
for bar, rate in zip(bars, rates):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.3,
            f'{rate:.1f}%', ha='center', fontweight='bold', fontsize=11)
ax.set_ylim(85, 102)
ax.set_ylabel('Retention Rate (%)')
ax.set_title('Customer Retention Rate by Segment (COVID Period)', fontweight='t
ax.axhline(y=100, color='gray', linestyle='--', alpha=0.3)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('chart_retention.png', dpi=150, bbox_inches='tight')
plt.show()
```

=== Customer Retention by Segment ===

Segment	2019 Customers	Retained	Lost	Retention Rate
Enterprise	12	12	0	100.0%
Large	15	15	0	100.0%
Medium	174	173	1	99.4%
Small	860	835	25	97.1%



✓ 14. Top 10 Customers by COVID Volume Growth

Beyond segment-level trends, we identified the top 10 individual customers who grew the most in absolute volume during the COVID period. Understanding which specific accounts drove the surge helps ABC Company prioritize relationship management — these are the accounts most worth nurturing with dedicated service agreements, capacity guarantees, or loyalty pricing. We also flag their customer group so leadership can see whether the growth leaders were concentrated in one segment or spread across the portfolio.

```
# Absolute volume growth per customer during COVID period
covid_period = df[(df['THE_YEAR']==2020) & (df['THE_WEEK']>=12)].groupby('FakeC
same_2019      = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']>=12)].groupby('FakeC

growth_df = pd.DataFrame({'CovidVol': covid_period, 'Vol2019': same_2019}).drop
growth_df['AbsGrowth'] = growth_df['CovidVol'] - growth_df['Vol2019']
growth_df['PctGrowth'] = (growth_df['AbsGrowth'] / growth_df['Vol2019'] * 100).
growth_df = growth_df.merge(vol_2019[['FakeCustomerID', 'CustomerGroup']], left_

top10 = growth_df.nlargest(10, 'AbsGrowth')[['FakeCustomerID', 'CustomerGroup', ']]
```

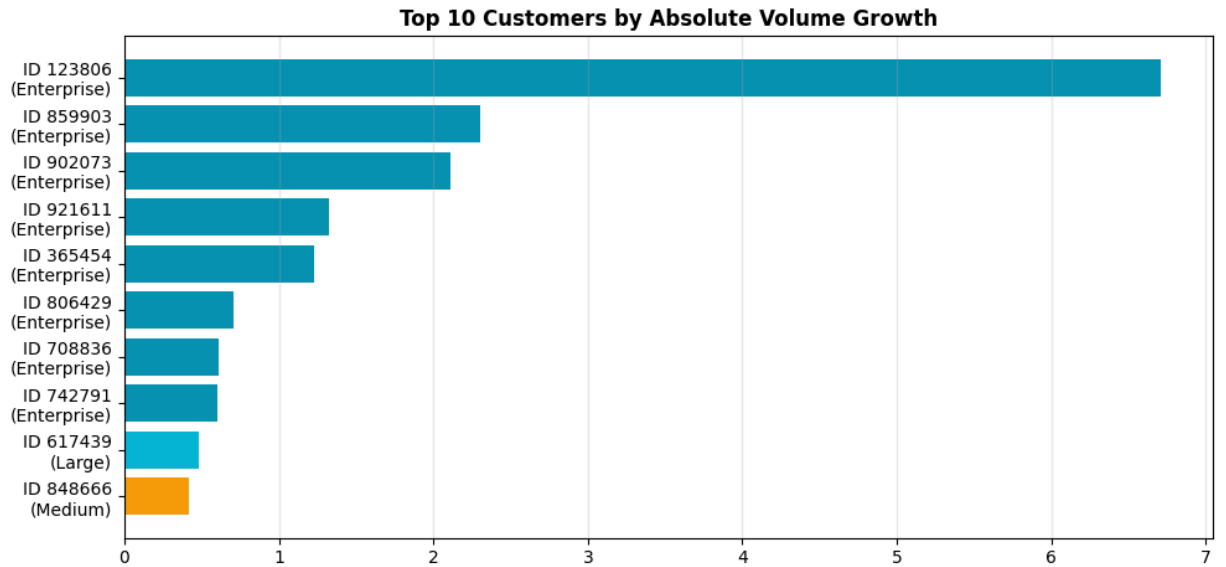
```
top10.columns = ['Customer ID','Group','2019 Vol (same period)','2020 COVID Vol']
top10 = top10.reset_index(drop=True)
top10.index += 1

print('=== Top 10 Customers by Absolute Volume Growth (COVID Period) ===')
print(top10.to_string())

# Horizontal bar chart
fig, ax = plt.subplots(figsize=(10, 5))
labels = [f"ID {row['Customer ID']}\n({row['Group']})" for _, row in top10.iter
values = top10['Absolute Growth'].values / 1e6
bar_color = ['#0891B2' if g == 'Enterprise' else '#06B6D4' if g == 'Large'
              else '#F59E0B' if g == 'Medium' else '#10B981'
              for g in top10['Group']]
ax.barh(labels[::-1], values[::-1], color=bar_color[::-1])
ax.set_xlabel('Additional Parcels During COVID Period (Millions)')
ax.set_title('Top 10 Customers by Absolute Volume Growth', fontweight='bold')
ax.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.savefig('chart_top10.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
=== Top 10 Customers by Absolute Volume Growth (COVID Period) ===
```

	Customer ID	Group	2019 Vol (same period)	2020 COVID Vol	Absolute Gr
1	123806	Enterprise	6908635	13618304.0	67096
2	859903	Enterprise	2067930	4371516.0	23035
3	902073	Enterprise	3344649	5459061.0	21144
4	921611	Enterprise	1177283	2505211.0	13279
5	365454	Enterprise	5146224	6370625.0	12244
6	806429	Enterprise	464464	1173415.0	7089
7	708836	Enterprise	703254	1313833.0	6105
8	742791	Enterprise	3331233	3932753.0	6015
9	617439	Large	312250	797429.0	4851
10	848666	Medium	56120	474839.0	4187



15. Summary & Recommendations

Key Findings

- 1. **COVID massively accelerated parcel volumes.** Total 2020 volume was 73.5M parcels — a **+39% increase** over 2019's 52.8M.
- 2. **The surge began at Week 14 (late March 2020)** — just after Canada's national emergency declaration (Week 11) and initial lockdowns (Week 12). By Week 16, volumes were up **+128% YoY**.
- 3. **Peak season 2020 was 26.5% larger than 2019** — driven by pandemic-accelerated e-commerce during the holiday period.

4. **Most customers grew during COVID.** Across all segments, 50–67% were classified as High Growth, reflecting the sector-wide e-commerce surge.
5. **Enterprise customers dominate volume** — accounting for 73.2% of 2020 parcels. Their strong growth (+38.3%) drove the bulk of revenue gains.
6. **Small customers showed the highest growth rate** (+51.9% volume), suggesting SMBs rapidly adopted e-commerce in response to lockdowns.